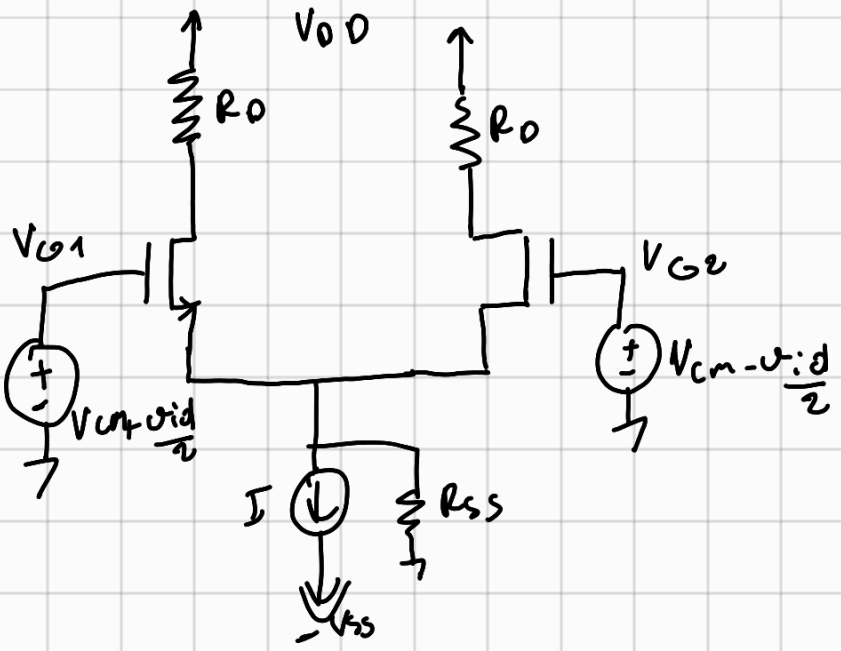
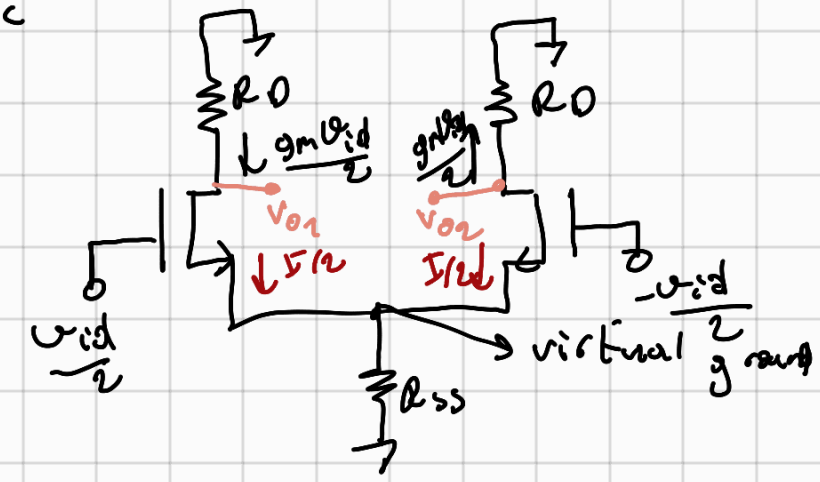
Small-Signal Operation of MOS Differential Pair



$V_{G1} = V_{CM} + \frac{v_{id}}{2}$

$V_{G2} = V_{CM} - \frac{v_{id}}{2}$

In ac



$v_{o1} = -\frac{g_m v_{id} R_D}{2}$

$v_{o2} = \frac{g_m v_{id} R_D}{2}$

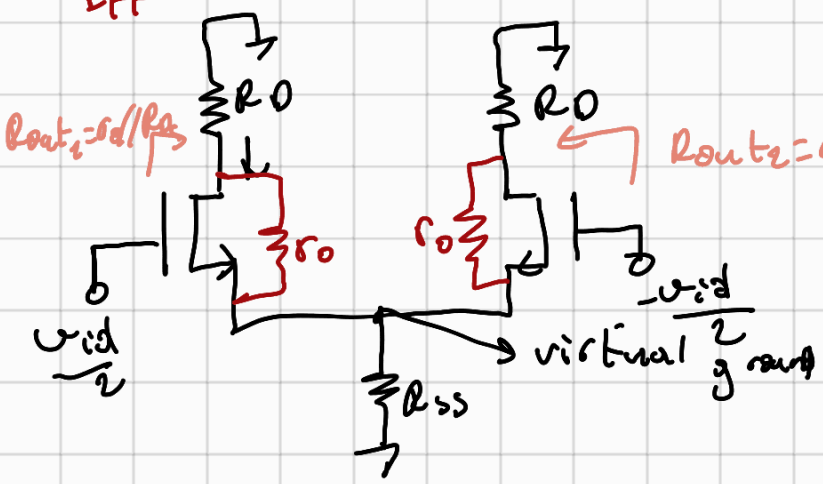
$|A_{single}| = |v_{o1}| = |v_{o2}| = g_m R_D/2$

$|A_{diff}| = \frac{|v_{o1} - v_{o2}|}{v_{ig}} = g_m R_D$

$g_m = \frac{2I_D}{V_{OV}} = \frac{I}{V_{OV}}$ ↳ $|A_{diff}| = \frac{I \cdot R_D}{V_{OV}}$

no channel length modulation ("r_o is ignored")

Effect of r_o

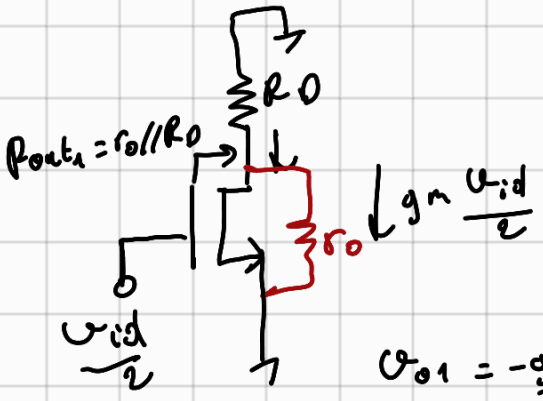


$$V_{o2} = |V_{o1}| = g_m \frac{v_{id}}{2} (r_o \| R_D)$$

$$|A_{single}| = \frac{1}{2} g_m (r_o \| R_D)$$

$$|A_{diff}| = g_m (r_o \| R_D)$$

The differential half circuit



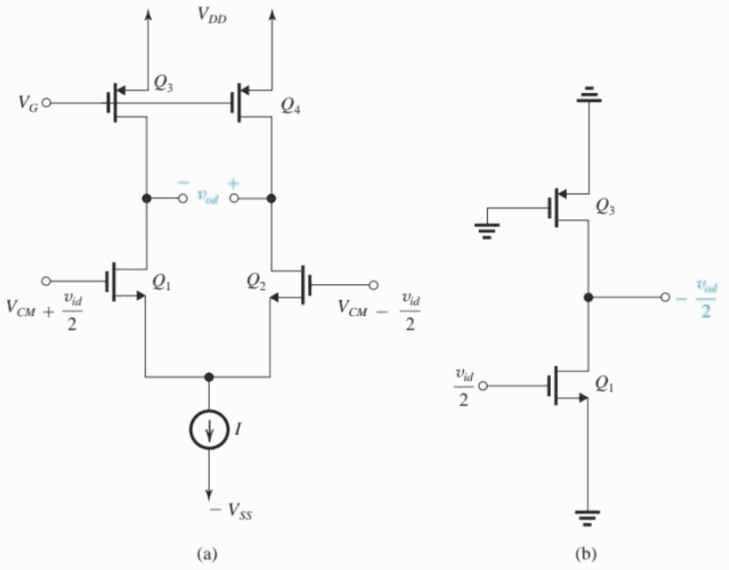
$$v_{o1} = -g_m \frac{v_{id}}{2} (R_D \| r_o)$$

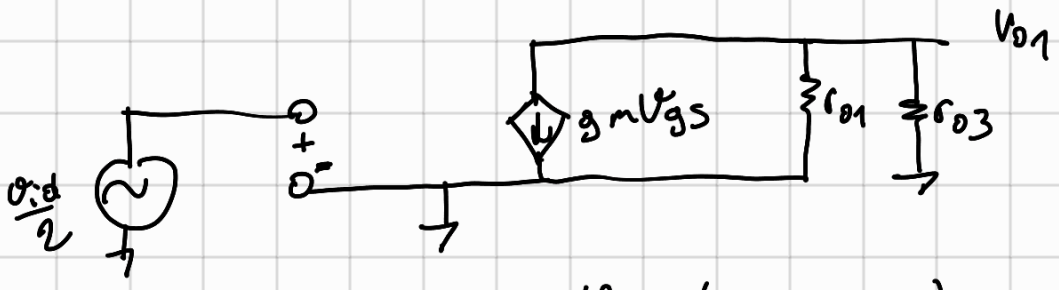
$$\frac{(v_{o1})}{(v_{id}/2)} = -g_m (R_D \| r_o)$$

$$\frac{v_{o1}}{v_{id}} = -\frac{1}{2} g_m (R_D \| r_o)$$

8.2.3

$$A_d \equiv \frac{v_{od}}{v_{id}} = g_{m1}(r_{o1} \parallel r_{o3})$$

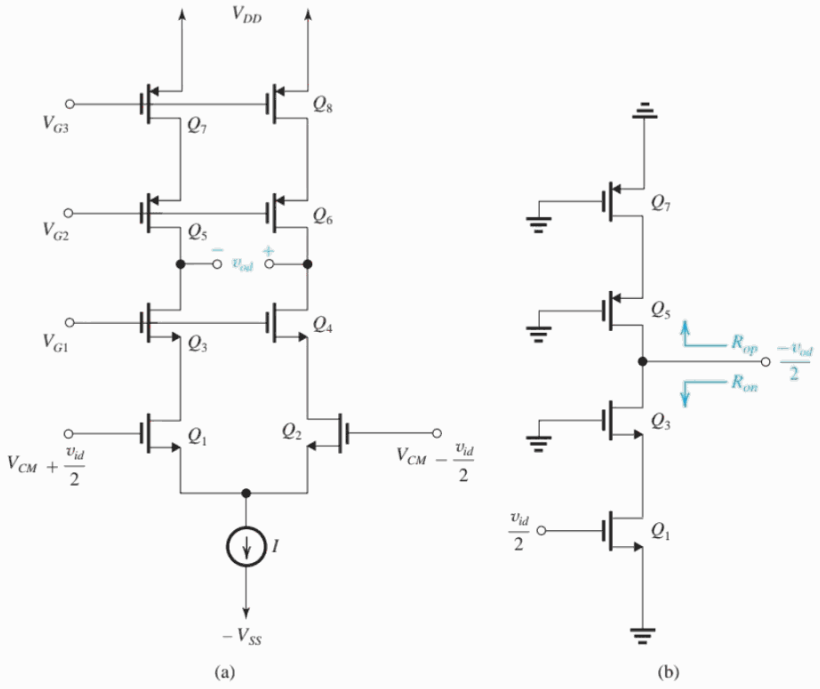




$$V_{o1} = -g_{mn} v_{gs1} (r_{o1} \parallel r_{o3})$$

$$\frac{V_{o1}}{v_{id}} = -\frac{1}{2} g_m (r_{o1} \parallel r_{o3})$$

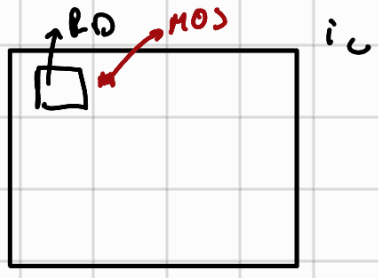
$$\frac{v_{o2} - v_{o1}}{v_{id}} = g_m (r_{o1} \parallel r_{o3})$$



$$R_{on} = R_{op} = g_m r_o^2$$

$$A_d = g_m \left(g_m r_o^2 \,//\, g_m r_o^2\right)$$

$$A_d = \frac{1}{2} g_m^2 r_o^2$$



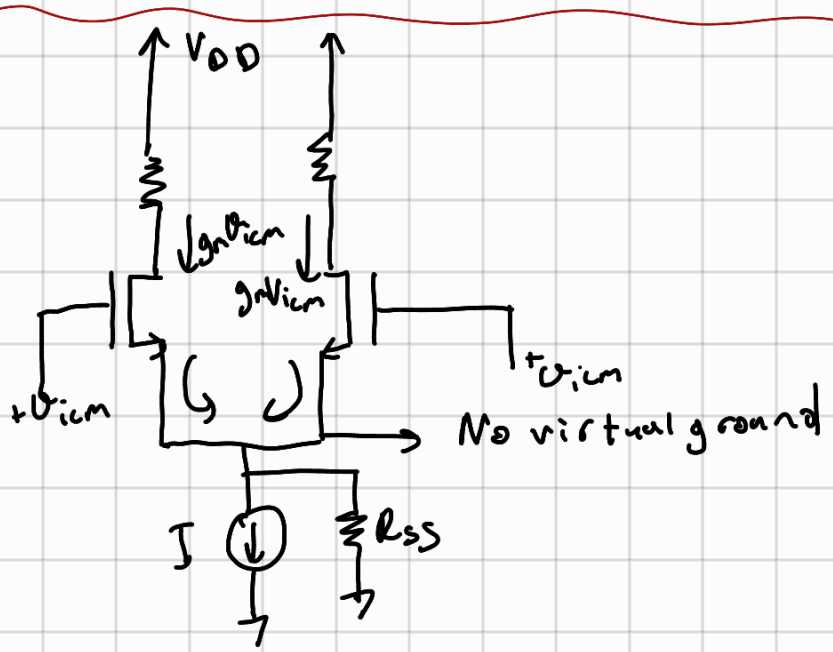
ic

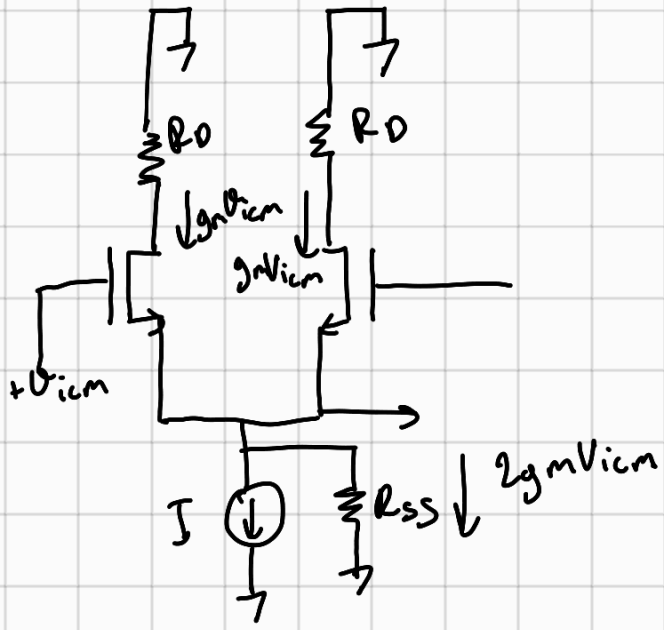
$$A_d \equiv \frac{v_{od}}{v_{id}} = g_{m1}(R_{on} \,\|\, R_{op})$$

$$R_{on} = (g_{m3} r_{o3}) r_{o1}$$

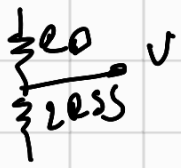
$$R_{op} = (g_{m5} r_{o5}) r_{o7}$$

Common-Mode Gain and Common-Mode Rejection Ratio





$R_D // 2R_{SS}$



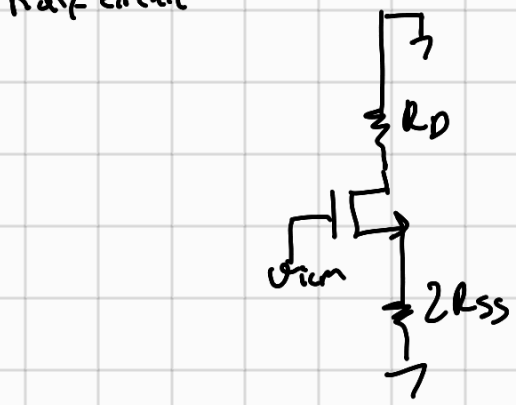
$$v_{o1} = -\frac{R_D}{2R_{SS}} v_{icm}$$

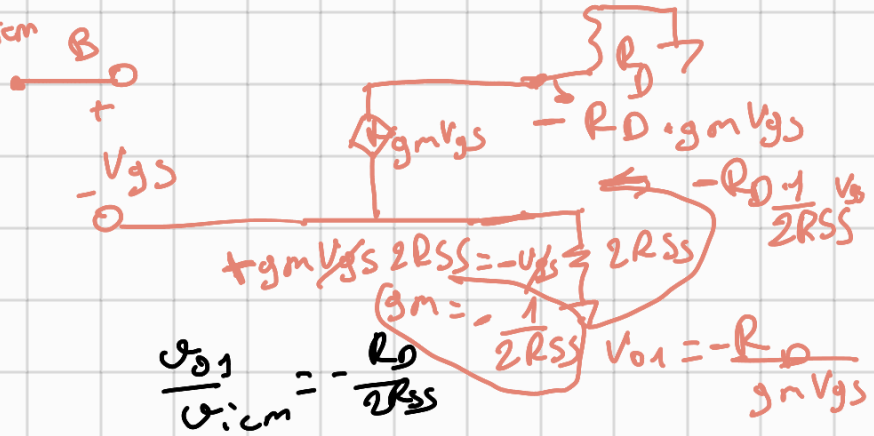
$$v_{o2} = \frac{-R_D}{2R_{SS}} v_{icm}$$

$$|A_{cm\,single}| \simeq \frac{R_D}{2R_{SS}}$$

$$|A_{cm\,dif}| = \emptyset$$

half circuit





$$\frac{v_{o1}}{v_{icm}} = -\frac{R_D}{2R_{SS}}$$

$$V_o = -2 g_m V_{gs} R_{SS}$$

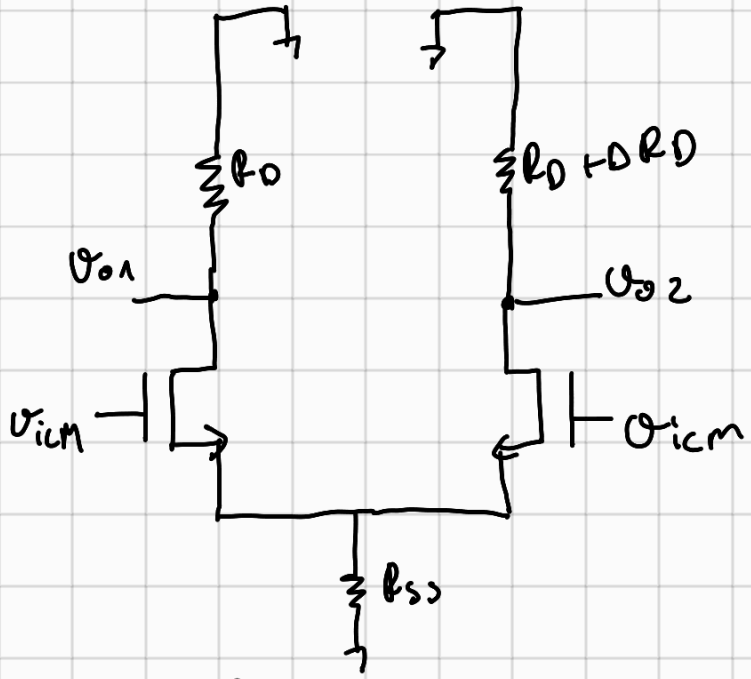
CMRR → ratio
- C → Common
- M → mode
- R R → Rejection

$$CMRR_{single} = \frac{|A_{single\,diff}|}{|A_{single\,common}|} \simeq \frac{\frac{1}{2} g_m R_D}{\frac{R_D}{2R_{SS}}} = g_m R_{SS}$$

in theory

$$CMRR_{diff} = \frac{|A_{diff\,diff}|}{|A_{diff\,cm}|} = \frac{g_m R_D}{0} = \infty$$

CMRR'in maximum olmasını isteriz.



$$v_{o1} = -\frac{R_D}{2R_{SS}}\, v_{icm}$$

$$v_{o2} = -\frac{R_{D1} + \Delta R_D}{2R_{SS}}\, v_{icm}$$

$$v_{o2} = -\frac{R_{D1} + \Delta R_D}{2R_{SS}}\, v_{icm}$$

$$v_{o2} - v_{o1} = \frac{\Delta R_D}{2R_{SS}}\, v_{icm}$$

$$A_{CM\,diff} = \frac{\Delta R_D}{2R_{SS}} = \left(\frac{R_D}{2R_{SS}}\right)\frac{\Delta R_D}{R_D}$$

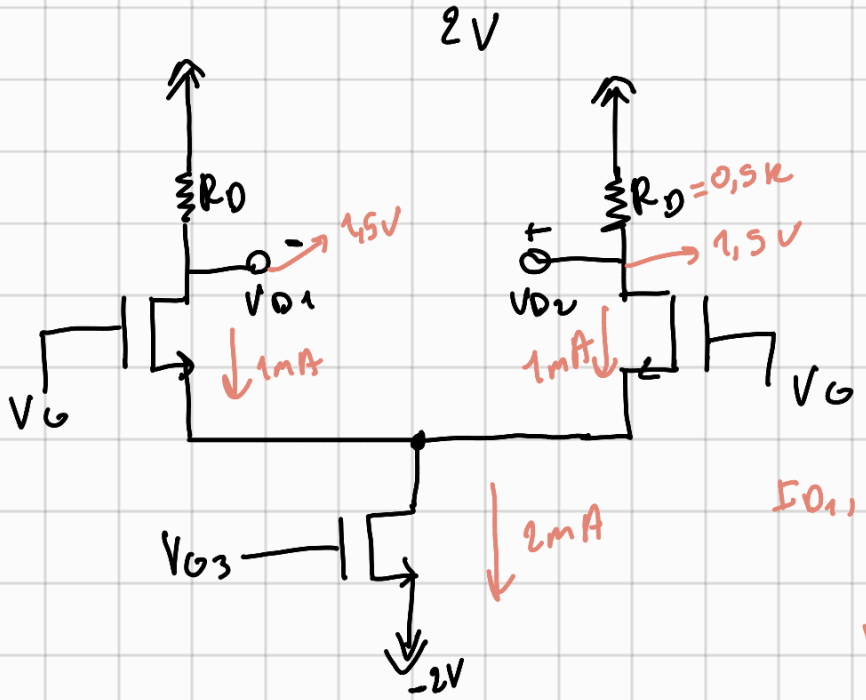
$$CMRR = \left|\frac{A_{diff\,diff}}{A_{cm\,diff}}\right| = \frac{g_m R_D}{\frac{\Delta R_D}{2R_{SS}}} = \frac{2 g_m R_{SS}}{\Delta R_D / R_D}$$

Effet of gm mismatch on CMRR $g_{m1} \neq g_{m2}$

$$CMRR_{diff} = \frac{2 g_m R_{SS}}{\Delta g_m / g_m}$$

$g_m \Rightarrow$ typical value

(1)



$I_{D1}, I_{D2} = 1mA$

$1 = \frac{1}{2} \cdot 8mA/V^2 (V_{OV})^2$

$V_{OV_{1,2}} = \pm 0,5V$

Calculate V_D, V_S, V_{DS} and V_{GS} if $I_{D3} = 2mA$, $R_D = 0,5k\Omega$ and $V_{OV3} = 0,5V$. Assume that Q_1 and Q_2 are identical with $\mu_n C_{ox}(W/L) = 8mA/V^2$

$V_{tn} = 0,5V$

$\lambda = 0$

$V_{GS1} - V_{tn} = 0,5V$

$0 - V_S - 0,5 = 0,5V$

$\boxed{V_S = -1V}$

$V_{DS} = 1,5 - (-1) = 2,5V$

$V_{GS} = 1V$

In saturation

if $\lambda \neq 0$ $\lambda = 0,1$

$I_{D1,2} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{OV})^2 (1 + \lambda V_{DS1})$

$1mA = \frac{1}{2} \cdot 8 \cdot (V_{OV})^2 \cdot (1 + \lambda V_{DS1})$

$(V_{OV})^2 (1 + \lambda V_{DS1}) = 0,25$

$V_{DS} = V_D - V_S = V_D + (V_{GS} - V_G)$

$= V_D + (V_{GS} - V_{Tn} - V_G) + V_{tn}$

$$V_{DS} = \cancel{V_0}^{1,5} + (V_{ov} - \cancel{V_0}^{0}) + \cancel{V_{tn}}^{0,5}$$

$$V_{DS} = V_{ov} + 2$$

$$(V_{ov})^2 (1 + 0,1 (V_{ov} + 2)) = 0,25$$

$V_{ov} = x$ olsun

$$x^2 (1 + 0,1x + 0,2) = 0,25$$

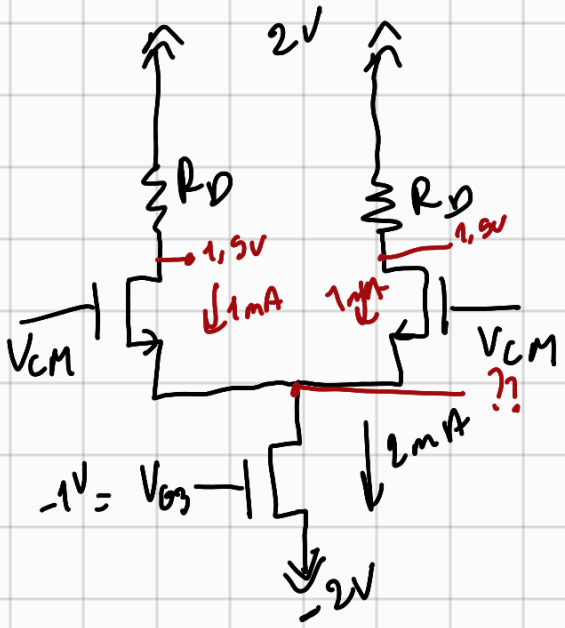
$$x^2 (1,2 + 0,1x) = 0,25$$

$V_{ov} = 0,448\ V$

$V_{GS1,2} = 0,948V$

$V_{S1,2} = -0,948V$

Common mode Range



What is the input common mode range? Q_1 and Q_2 are identical

$R_D = 0,5k\Omega$

$\mu_n C_{ox} W/L = 8\ mA/V^2$

$V_{t1,2,3} = 0,5V$

$V_{GS3} = -1V$

$$1 = \frac{1}{2} 8mA/V^2 (V_{ov})^2$$

$V_{GS} - V_{tn} = 0,5$

$V_{GS} = 1V$

$V_D = 2 - 0,5\,mA$

$= 1,5V$

$V_{GS3} = -1 - (-2) = 1V$

$V_{ov} = 1 - 0,5 = 0,5V$

$V_{DS3} \geq V_{ov3}$ satkarak

$V_{DS3} \geq 0,5V$

$V_{D3} \geq -1,5V$

$V_{CM} - (-1,5) - 0,5 \geq 0$

$\boxed{V_{CM} \geq -1V}$

Triodtan kaçınması lazım üsttekinin

$V_{DS} \geq V_{GS} - V_t$

$V_D \geq V_{CM} - 0,5$

$V_D + 0,5 \geq V_{CM}$

$2 \geq V_{CM}$

$-1V \leq V_{cm} \leq 2V$

input cm range //

3) A MOS differential pair is operated at a bias current(I) of 0,4 mA.

If $\mu_n C_{ox} = 0,2 mA/V^2$, find the required values of $(\frac{W}{L})$ and the resulting g_m if the Mosfets are operated at V_{OV}: 0,2V, 0,3V and 0,4V. for each value give the max v_{id}

$I = 0,4mA \rightarrow I_{D1} = 0,2mA$, $I_{D2} = 0,2mA$

$0,2mA = 1/2 \mu_n C_{ox} (\frac{W}{L}) (V_{ov})^2$

$2 = (\frac{W}{L}) / (V_{OV})^2$

for $V_{OV} = 0,2V$

$(\frac{W}{L}) = \frac{2}{(0,2)^2} = 50 \Rightarrow W = 50\mu m$, $L = 1\mu m$

$g_m = \frac{2I_D}{V_{OV}} = \frac{2 \cdot 0,2}{0,2} = 2mS$

max $(v_{id}) = ?$

$-\sqrt{2} V_{OV} \leq v_{id} \leq \sqrt{2} V_{OV}$

$\simeq 0,2 \cdot 0,8 \simeq 250mV$ //